

The empirical recurrence for OEIS A259425: recovering a conjecture that is not written down

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Abstract

OEIS A259425 records “Empirical recurrence of order 62”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 72$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A259425 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with each 2×2 subblock having clockwise pattern 0000 0001 0011 or 0111.”. It has offset 1 and begins

5116, 244228, 11155819, 516880938, 23840819225, 1101527291223, 50866278137234, 2349427850331898, ...

The entry states:

Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:51:26, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 129$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 72$ of the 129, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 144$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	55	c_{17}	123681409026174623	c_{33}	−18948350963490287950774	c_{49}	−189998085
c_2	431	c_{18}	−471363630657879071	c_{34}	−55878371263193246645790	c_{50}	−89327603
c_3	−43793	c_{19}	−1389712255814911700	c_{35}	35568688607421932962310	c_{51}	211965920
c_4	−23569	c_{20}	6026948664771309841	c_{36}	62870388741113733814425	c_{52}	39459166
c_5	13144286	c_{21}	10782702490915434168	c_{37}	−46450277733030166517653	c_{53}	−15448246
c_6	−16155243	c_{22}	−55348879194784448972	c_{38}	−52111378689663296106714	c_{54}	−1307696
c_7	−1986919003	c_{23}	−56315179764335544894	c_{39}	43283203713426134130059	c_{55}	70894642
c_8	3863339919	c_{24}	370214710418627253204	c_{40}	31458807693940311175188	c_{56}	8965966
c_9	171913764370	c_{25}	180513188090113596477	c_{41}	−28993166021330798472024	c_{57}	−194051
c_{10}	−410006555510	c_{26}	−1823089310560734064866	c_{42}	−13609056399405057611821	c_{58}	−51624
c_{11}	−9223398179643	c_{27}	−206776179123756613805	c_{43}	13936759185460947764796	c_{59}	284884
c_{12}	25261297930246	c_{28}	6660378410402078606245	c_{44}	4123193465281153460547	c_{60}	127203
c_{13}	323019352234946	c_{29}	−1050860467383863511474	c_{45}	−4769106894913576927925	c_{61}	−1678
c_{14}	−991239939573646	c_{30}	−18135836286733534173832	c_{46}	−846598694052440054808	c_{62}	−1029
c_{15}	−7626002939224075	c_{31}	6520115194938276946435	c_{47}	1146358585250229758862		
c_{16}	26018989779271437	c_{32}	36867422445836883382534	c_{48}	112226732854936653061		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62 + 4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

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